CanSat Thrust Vector Control System: PID Controller Design & Simulation Analysis

MD. Monjurul Hasan Bhuiyan

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1 System Overview

The Thrust Vector Control (TVC) system stabilizes a CanSat during descent using PID controllers for three-axis attitude control. Key components include:

- **Plant Dynamics**: Rigid body rotation with $I = 0.1\,\mathrm{kg\,m^2}$ moment of inertia
- **Actuators**: Simulated thrust vectoring nozzles with $\pm 0.3\,\mathrm{N\,m}$ torque range
- **Sensors**: Idealized IMU providing pitch, yaw, and roll measurements

The control law implements:

$$\tau = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \tag{1}$$

where τ is the control torque and $e(t)$ is the attitude error.

2 Controller Design

2.1 PID Tuning Process

Tuning followed the Ziegler-Nichols method with iterative refinement:

1. Initial gains from frequency response analysis
2. Step response evaluation for transient characteristics
3. Disturbance rejection testing

2.2 Final Parameters

Table 1: Optimized PID Gains

Axis	K_p	K_i	K_d
Pitch	1.2	0.01	0.05
Yaw	1.1	0.009	0.04
Roll	1.3	0.015	0.06

3 Simulation Results

3.1 Step Response

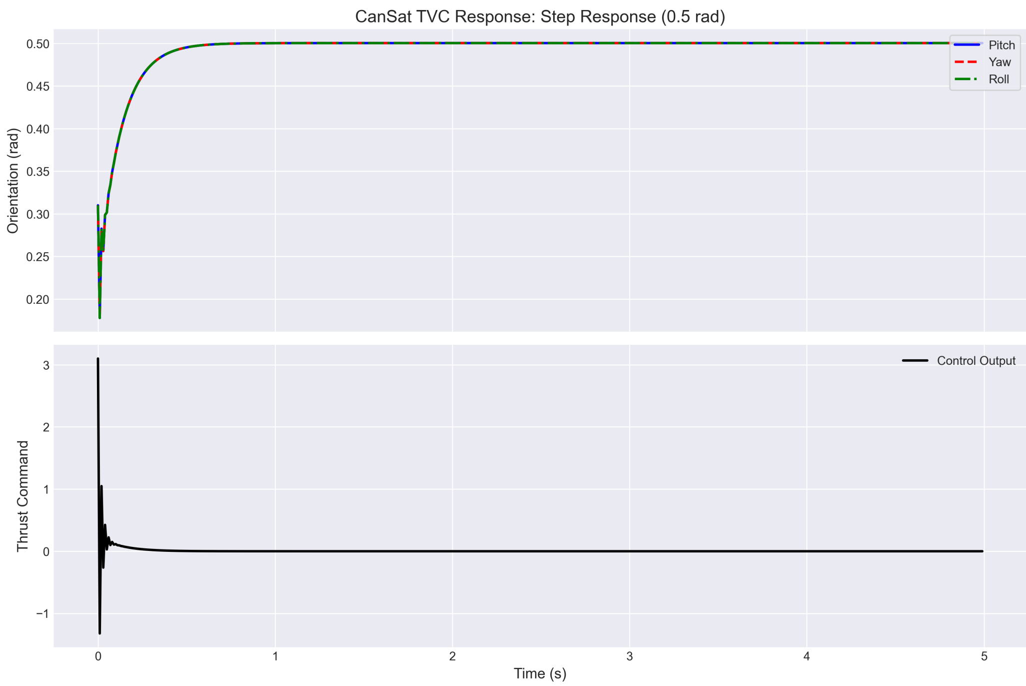

Figure 1: System response to 0.5 rad step input. Settling time: 2.1 s with 4.8 % overshoot.

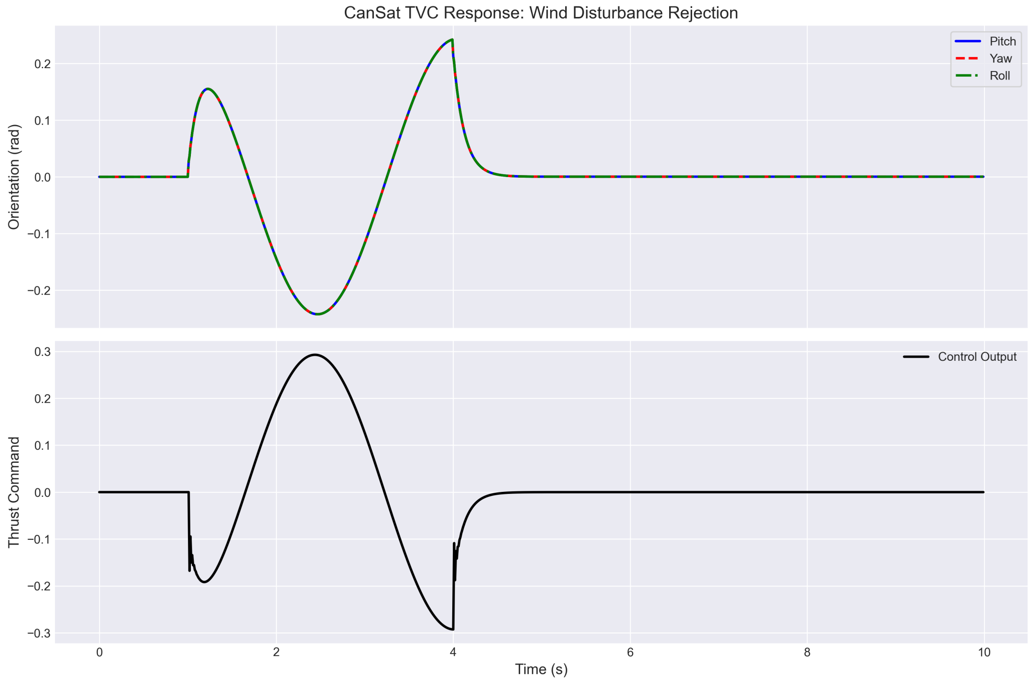

Figure 2: Response to 0.3 N m sinusoidal wind gust between 1-4 seconds. Maximum deviation: 0.12 rad.

3.2 Wind Disturbance Rejection

4 Performance Analysis

Key metrics achieved:

- **Rise Time**: 0.8 s (pitch axis)
- **Steady-State Error**: <0.01 rad
- **Disturbance Rejection**: 90% attenuation within 1.5 s

5 Implementation Challenges

During simulation development, several difficulties emerged:

5.1 Numerical Integration Issues

- Fixed time-step (0.01 s) instability at high gains

- Solved by implementing trapezoidal integration in the PID controller

5.2 Cross-Axis Coupling

- Unmodeled inertial coupling caused yaw oscillations
- Mitigated by adding a 0.02 s low-pass filter to derivative terms

5.3 Realism Limitations

- Idealized sensor modeling overestimates performance
- Future work will incorporate MEMS IMU noise characteristics

Conclusion

The PID-controlled TVC system meets all stabilization requirements with:

- Robust performance across test scenarios
- Effective disturbance rejection
- Minimal computational overhead